

Douglas D'Avila

QA Architect • Software Development Engineer In Test • AI-Driven Solutions

Porto Alegre, RS, Brazil | douglas.odavila@gmail.com | +55 51 98127-0800 |
linkedin.com/in/douglasottodavila | github.com/DouglasOttoDavila

Professional Summary

AI-driven QA Architect and SDET with 5+ years of experience designing automation strategies, building scalable test frameworks, and delivering SDLC-enhancement tools for global eCommerce and enterprise platforms. Combines hands-on engineering in Python, JavaScript/TypeScript, Playwright, Selenium/WebdriverIO, API testing, CI/CD, and performance testing with leadership in Agile environments. Known for turning QA bottlenecks into practical automation, AI-assisted analysis, and reusable workflows that improve coverage, speed, and engineering decision-making.

Key Differentiators

- Specialized in AI-powered QA solutions, including test case generation, requirements validation, defect analysis, and visual regression interpretation.
- Strong SDET foundation across UI, API, CI/CD, performance, and test management workflows, with practical delivery experience rather than tool-only exposure.
- Proven automation leadership, including mentoring QA engineers/SDETs and reducing manual intervention in pipeline-driven testing by 40%.
- Able to translate business requirements into test strategy, automation architecture, executable test cases, and measurable quality outcomes.

Technical Skills

Languages

Python, JavaScript, TypeScript, Java

API & Integration Testing

Postman, REST Assured, REST APIs, service-level validation

AI & LLM Tools

Azure AI, OpenAI, Gemini, AI-assisted SDLC workflows

Test Management

TestRail, Zephyr, traceability, coverage analysis

Test Automation

Playwright, Selenium, Cypress, WebdriverIO, Robot Framework

CI/CD & DevOps

Jenkins, GitHub Actions, Bitbucket, Docker, pipeline automation

Performance Testing

JMeter, load and performance validation practices

Collaboration & Delivery

Jira, MantisBT, Scrum, Kanban, cross-functional leadership

Professional Experience

Software Development Engineer in Test (SDET)

Object Edge Inc. | Oct 2023 - Present

- Develop custom AI-powered applications that enhance SDLC processes, especially around test generation, requirements review, defect analysis, and automation decision support.
- Build and maintain intelligent QA automation solutions that combine LLM capabilities with existing engineering workflows and enterprise project context.
- Drive innovation by integrating AI-driven methodologies into test automation, CI/CD workflows, and quality engineering practices.
- Collaborate with cross-functional teams to deploy AI-enhanced automation solutions across projects while keeping outputs practical, reviewable, and aligned with delivery needs.
- Contribute to reusable QA tools and proof-of-concepts that improve how teams evaluate requirements, generate test coverage, and analyze testing outcomes.

QA Automation Lead

Object Edge Inc. | Feb 2020 - Oct 2023

- Spearheaded automation testing strategies across multiple enterprise eCommerce projects, aligning automation scope with risk, release needs, and business priorities.
- Established and refined CI/CD pipeline automation, reducing manual intervention by 40% and improving repeatability of regression execution.
- Led and mentored QA engineers and SDETs in test automation best practices, framework usage, maintainability, and reliable delivery habits.
- Developed and maintained automated test suites for UI and API validation, improving regression confidence across release cycles.
- Implemented API testing strategies and executed performance testing initiatives to validate reliability, integration behavior, and system readiness.

Selected AI & Automation Work

- AI test case generation and recommendation: designed workflows that use story context and existing test assets to suggest or generate more relevant coverage.
- Requirements review support: built AI-assisted analysis flows to evaluate user stories against quality, completeness, and testability criteria.
- AI-assisted defect and result analysis: explored LLM-based summarization and classification to speed up investigation of failures and testing outcomes.
- Visual regression intelligence: worked on concepts that combine pixel-level diffing with semantic AI review to separate meaningful UI regressions from harmless visual noise.
- Workflow-based QA tooling: contributed to the evolution from custom Python/Flask utilities toward more modular AI and automation workflows.

Career Impact Themes

Automation Architecture

Designs maintainable automation solutions for UI, API, and CI/CD workflows, with emphasis on reusability, clarity, and long-term maintainability.

Quality Leadership

Mentors QA professionals, raises engineering standards, and helps teams adopt automation practices without losing business context.

AI-Driven QA Innovation

Applies LLMs and AI services to practical QA problems such as test generation, requirements analysis, visual review, and defect triage.

Enterprise Delivery

Works across global eCommerce and enterprise environments where reliability, scalability, and stakeholder communication are essential.

Education

Technology Degree, System Analysis & Development - UniRitter | 2021 - 2023

Certifications

- AWS Academy Graduate - AWS Academy Cloud Foundations - Amazon Web Services | May 2023
- Red Hat OpenShift I: Containers & Kubernetes (DO180) - Red Hat | May 2023
- JavaScript Algorithms and Data Structures - freeCodeCamp | Dec 2022
- API Test Automation with Postman - Udemy | Mar 2022
- Cypress Discovery (Cypress, JavaScript, Node.js, GitHub Actions) - QASkills | Mar 2022
- Developer Accessibility Fundamentals I - eSENTIAL Accessibility | Aug 2021

Languages

Brazilian Portuguese: Native | English: C2 Proficient

Additional Positioning

Target roles: QA Architect, SDET Lead, QA Automation Architect, AI-Driven QA Engineer, Test Automation Lead, and quality engineering roles focused on scalable automation and SDLC modernization.